



S-SQUARED FPGA Development Kit

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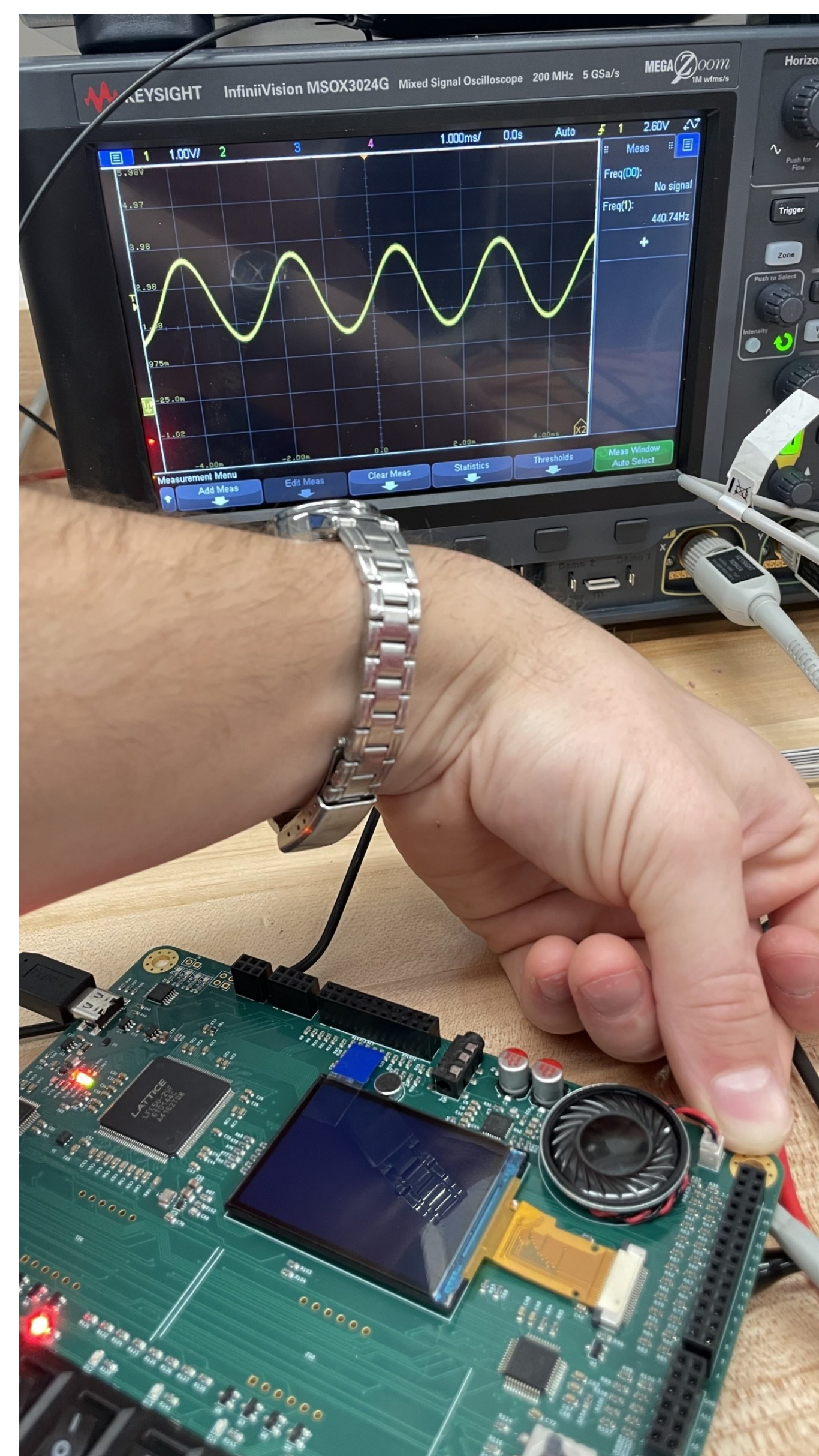
Abstract

There is a lack of student oriented Field-Programmable Gate Array (FPGA) evaluation boards on the market. Available products are, on the whole, expensive and not well designed for learning environments. The S-Squared FPGA is a low cost, versatile, and durable FPGA Development Kit designed to be used in college lab environments. The S-Squared FPGA is designed to be used and maintained at universities.

Background

Evaluation boards are crucial tools for the development of embedded systems.

There are a range of microcontroller development kits available to satisfy the diverse needs of consumers. The range of available FPGA evaluation boards is more constrained. Currently available development kits for FPGAs fall into the categories of very simple or very expensive

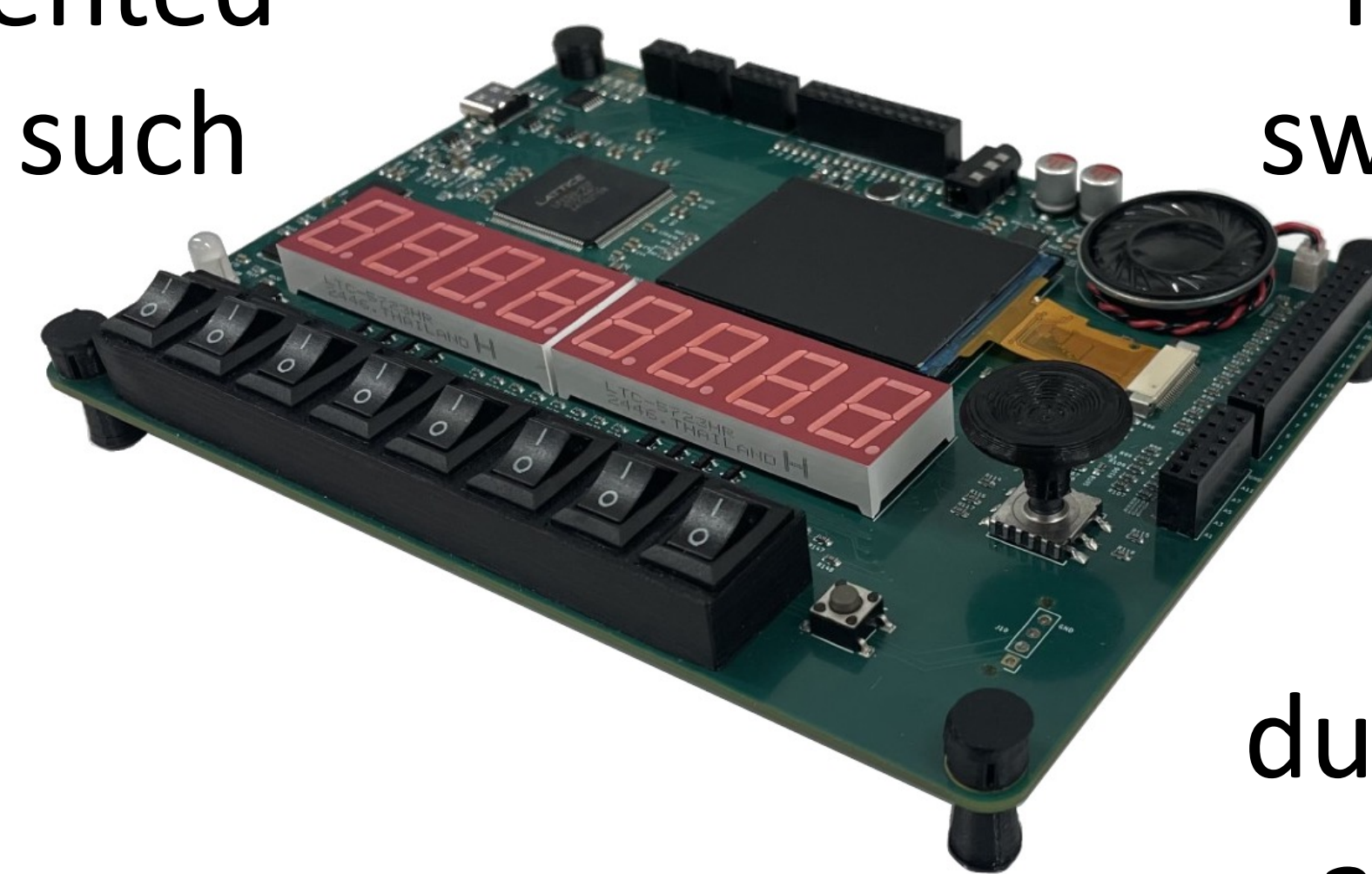


Methods and Materials

- The S² FPGA is priced competitively with the simpler models while providing more features than many expensive options.
- The core chips on the S² FPGA are designed with protection from accidental connections and erroneous code in mind.

S² FPGA

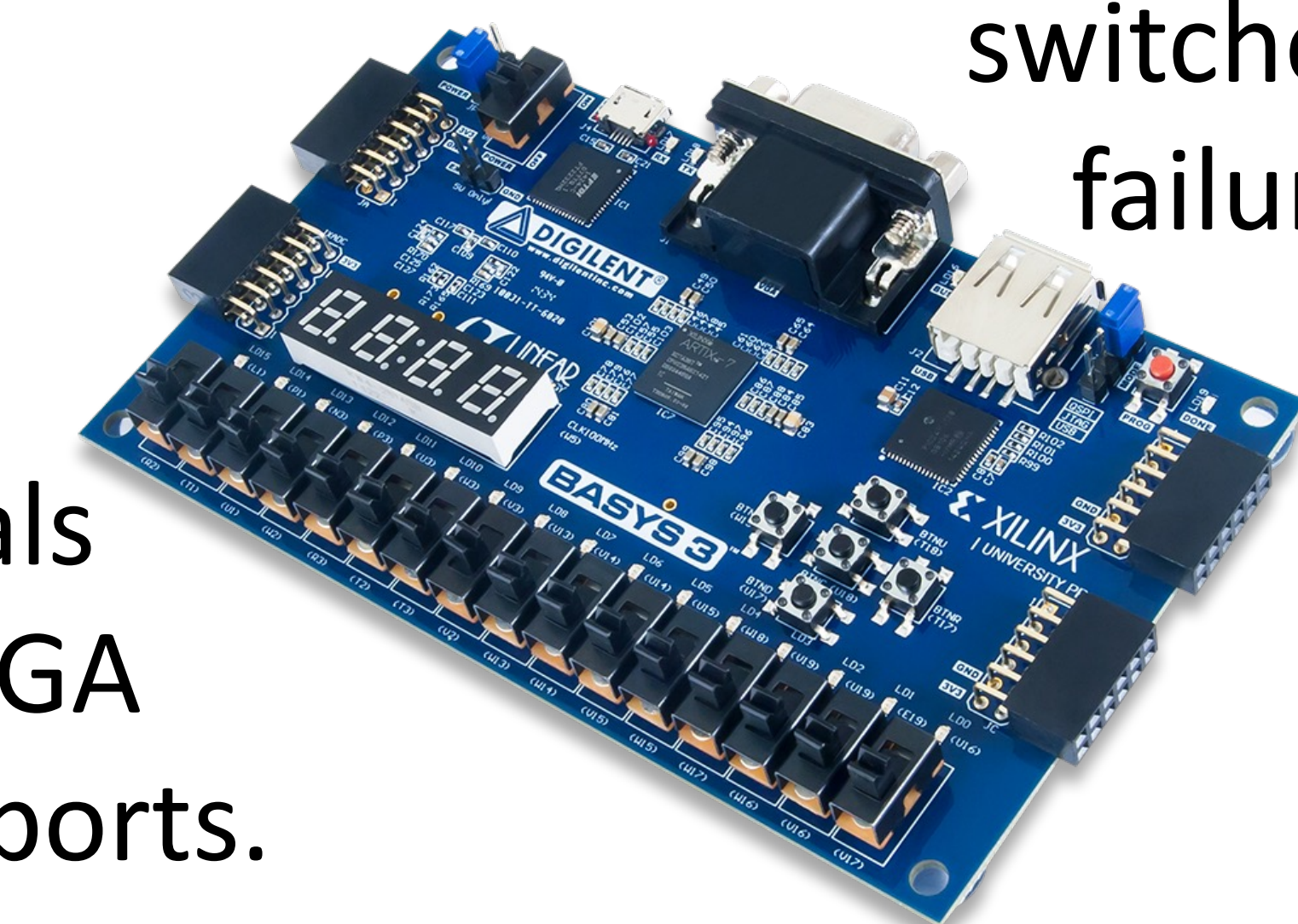
Student oriented peripherals such as joystick, screen, and speaker.



High quality switches, and large 7-segment displays for durability and accessibility.

Basys 3

Industry focused peripherals such as VGA and USB ports.



Low quality switches prone to failure, and tiny 7-segment displays.

Results

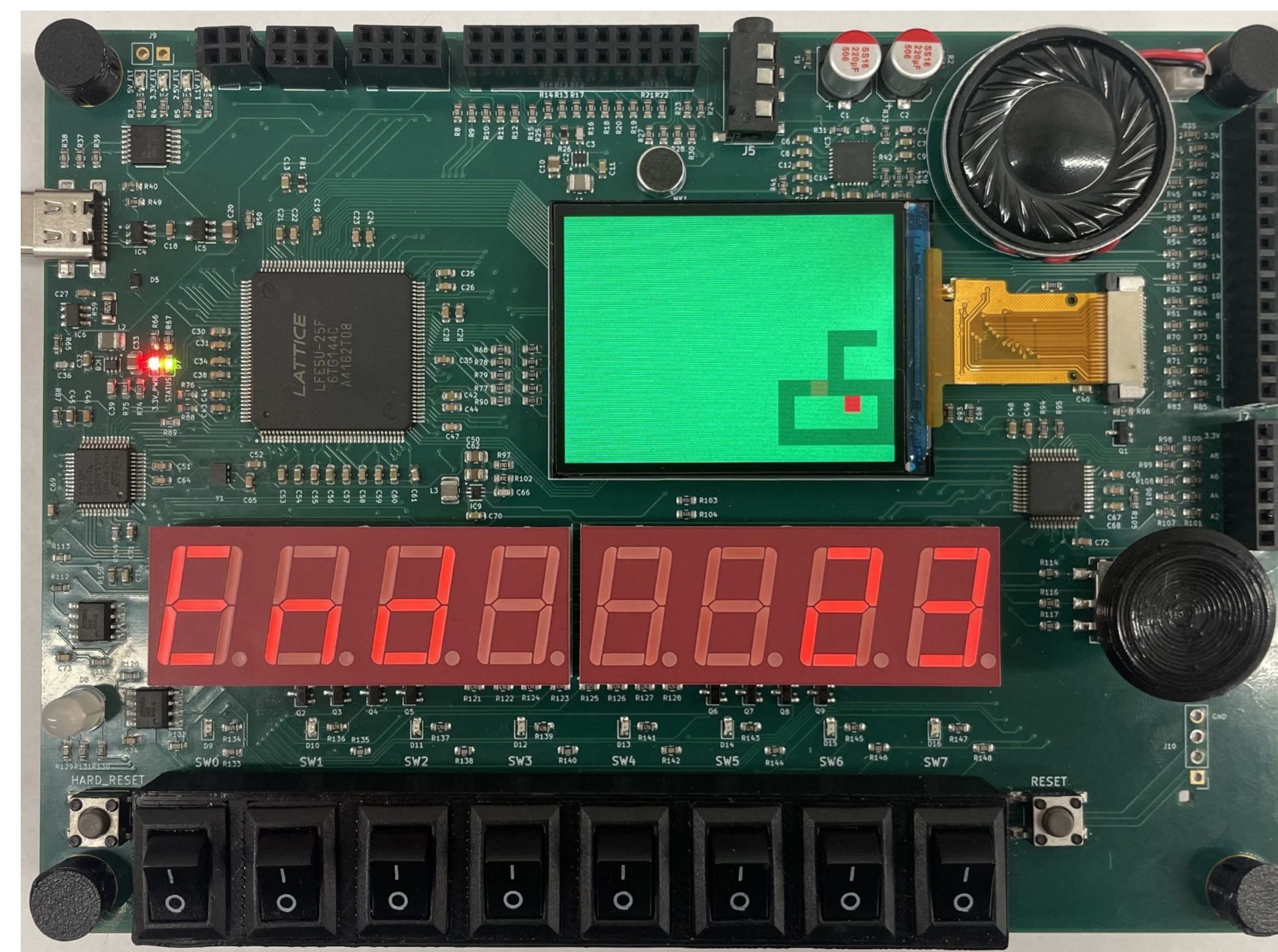


Fig. 1: S² FPGA displaying the game over screen of the snake demo program.

- The S² FPGA provides a valuable tool to anyone looking to get started using FPGAs.
- A simple introductory lab has been tested and approved of by a volunteer junior.
- GPIO pins have all been shorted, and the S² FPGA continued to function properly.

Future Direction

- We are in the process of closing a sale of 24 boards to WWU to be used for EECE 333.
- We also plan to contact several other universities including UW, EWU, and WSU to see if they have any interest in purchasing any boards.



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Capstone Project
Electrical & Computer Engineering Program